

M002 Week 7 — Pistons

Topic: Pistons and Mazes · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

A piston pushes or pulls a block when it gets a redstone signal. A **sticky piston** can pull a block back, opening a path — but only while the signal is ON. A lever sends that signal. Your agent flips the lever to power the piston, then crosses the gap it opens.

1 · Predict

Read each block. Write what happens before you imagine it.

```
agent.move(FORWARD, 1)
agent.interact(...)
agent.move(FORWARD, 1)
```

The agent walks to a lever, flips it, and a sticky piston pulls a block back. Can the agent move forward now?


```
if not agent.detect(BLOCK, FORWARD):
    agent.interact(...)
else:
    agent.move(FORWARD, 1)
```

What does the agent do at a gap? What does it do on solid ground?

2 · Trace 🐾

```
while True:
    if not agent.detect(BLOCK, FORWARD):
        agent.interact(...)
        agent.move(FORWARD, 1)
    else:
        agent.move(FORWARD, 1)
```

Describe step by step what the agent does as it moves through the maze.

3 · Spot and Fix Bugs 🐛

Each block is broken. Read the goal, write the fix, then explain it.

Bug A — Power the piston **first**, then cross the gap it opens.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug B — `interact` should run only when there is a **gap** ahead. This flips a lever on every step.

```
while True:
    agent.interact(...)
    agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug C — A sticky piston holds the bridge **only while the lever is ON**. The second `interact` switches it OFF and the bridge vanishes. Cross first.

```
agent.interact(...)
agent.move(FORWARD, 1)
agent.interact(...)
agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong? Why does your fix work?

4 · Write Full Code 🖋️

Write code that walks the agent to the lever, flips it so the piston opens the path, crosses while the path is open, and reaches the goal. Use a chat trigger to start it.

```
def cross_gap(name, message):
    pass # your code here

player.on_chat("go", cross_gap)
```

5 · Build It

Open your homework world. The maze has a gap (or blocked path) and a piston that opens it when powered. Run your code so the agent reaches the goal.

Send a photo or video of the agent at the end. Then finish these lines — 4 to 6 sentences.

First, I walked the agent to ...

I powered the piston by ...

Once the path opened, the agent ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

6 · Plan Your Walkthrough Video 🎥

Take a phone video while the agent runs the maze. Talk through it like you are teaching someone new. Use these words: **piston, redstone, signal, gap, bridge.**

1. Show the gap and the piston.
2. Run your code and show the agent powering the piston and crossing.
3. Read your code out loud and say which line opens the path.
4. Show one bug you hit and how you fixed it.

Write what you will say. Plan it here before you record.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.